



Python Programming Fundamentals

Mini CRUD with Dictionaries

Programming Fundamentals Team

SETU

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Outline

1. **CRUD**
2. **Program Structure**
3. **Add Movie**
4. **View Movies**
5. **Main Loop**



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1. CRUD

CRUD stands for:

- **Create** — add a new record
- **Read** — view records
- **Update** — change a record
- **Delete** — remove a record



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- **Create** — add a new record
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CRUD is the backbone of many applications.



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2. Program Structure

```
movies = []

def show_menu():
    print("1. Add movie")
    print("2. View movies")
    print("3. Search movie")
    print("4. Exit")
```



2. Program Structure

```
movies = []

def show_menu():
    print("1. Add movie")
    print("2. View movies")
    print("3. Search movie")
    print("4. Exit")
```

A menu lets the user repeatedly choose an action.



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3. Add Movie

```
def add_movie(movies):  
    movie = {  
        "title": input("Title: "),  
        "year": int(input("Year: ")),  
        "rating": int(input("Rating: "))  
    }  
    movies.append(movie)  
    print("Movie added")
```



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4. View Movies

```
def view_movies(movies):  
    if len(movies) == 0:  
        print("No movies yet")  
    else:  
        for movie in movies:  
            print(f"{movie['title']} ({movie['year']}) -  
{movie['rating']}/5")
```



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5. Main Loop

```
while True:
    show_menu()
    choice = input("Choose: ")

    if choice == "1":
        add_movie(movies)
    elif choice == "2":
        view_movies(movies)
    elif choice == "4":
        break
    else:
        print("Invalid choice")
```



5. Main Loop

```
while True:
    show_menu()
    choice = input("Choose: ")

    if choice == "1":
        add_movie(movies)
    elif choice == "2":
        view_movies(movies)
    elif choice == "4":
        break
    else:
        print("Invalid choice")
```

This is a complete application pattern.

Thanks for Watching - Any questions?

